# Shaon Kanti Dhar

*Project Leader* | *Samsung R&D Institute Bangladesh*

Email: shaonkantidhar07@gmail.com · LinkedIn: linkedin.com/in/shaon-kanti-dhar
Portfolio: https://shawonnstu.github.io/ · Dhaka, Bangladesh

## RESEARCH INTERESTS

Security and Privacy of Connected Devices, AI Agents and LLM Systems, and On-device AI (exploring speaker verification, audio classification, agentic workflows and retrieval-augmented generation)

## EDUCATIONAL BACKGROUND

**Noakhali Science and Technology University (NSTU), Noakhali, Bangladesh**
B.Sc. in Information and Communication Engineering **2015 - 2019**

- CGPA: 3.51/4.00
- Final-year project: Travel Assistant, an Android app with automated image screening

**Feni Government College, Feni, Bangladesh**
Higher Secondary Certificate (HSC), Science **2014**

- GPA: 4.80/5.00

**Feni Government Pilot High School, Feni, Bangladesh**
Secondary School Certificate (SSC), Science **2012**

- GPA: 5.00/5.00 (Golden)

## RESEARCH EXPERIENCE

**Security for Bluetooth Multipoint Earbuds**
***Samsung R&D Institute Bangladesh · Sole Researcher*** **Dec 2024**

- Identified how shared or multipoint earbuds can leak calls and private notifications to the wrong listener
- Proposed per-earbud trust levels from on-device speaker verification and contextual signals (wear sensors, connection state, location)
- Designed an audio sensitivity classifier using rules plus ASR and NLP; audio is routed only to sufficiently trusted earbuds
- **Status:** Invention proposal, not granted

**Accidental-Touch Filtering for Wireless Earbuds**
***Samsung Electronics · Co-inventor*** **Patent**

- Method using sensor data to separate intentional touches from accidental ones on earbud touchpads
- Graded A1 by SRBD Patents · **Patent Link:** US20230229260A1

**Flicker Reduction in Multi-Camera Video Recording**
***Samsung Electronics · Co-inventor*** **Patent**

- Method that synchronizes camera settings to reduce light flicker when several cameras record at once

**Automated Screening of User-Shared Images**
***Noakhali Science and Technology University · Undergraduate Project*** **Oct 2018 - May 2019**

- Android travel app that screens shared photos with Firebase ML Kit and Microsoft's content moderation API
- **Report Link:** PDF

---

## AI PROJECTS

**SmartThings Pro B2B AI Agent**
***Samsung R&D Institute Bangladesh · Architect*** **2025 - Present**

- Architected the AI agent for Samsung's B2B building-management platform and led its rollout to Starbucks

**Code Quality Agent**
***Samsung R&D Institute Bangladesh · Designer***

- One agent with three triggers (monthly schedule, GitHub webhooks, continuous review monitoring) sharing one state store
- Opens monthly pull requests that raise code-quality scores (SAM, TEM, CQM), fixes review comments and escalates after three rounds
- A quality gate requires a CQM score of 5.00 with no critical or major issues; results are published to a dashboard after merge

---

## EXPERIENCE

**Project Leader**
***Samsung R&D Institute Bangladesh, Dhaka, Bangladesh*** **Dec 2025 - Present**

- Leading a team of 12 engineers on SmartThings Pro, Samsung's B2B IoT and building-management platform
  - Used by Starbucks and installed at Kumho Resort, South Korea
  - Architected the SmartThings Pro B2B AI Agent and rolled it out to Starbucks
  - Set up an AI-assisted development environment using LLMs, RAG and tool integration (MCP)
- Own architecture, API design, reliability and web performance: request parallelization, client-side caching, query orchestration, list virtualization and real-time updates

**Lead Engineer**
***Samsung R&D Institute Bangladesh, Dhaka, Bangladesh*** **Mar 2025 - Dec 2025**

- Led the Lounge and Virtual Studio showroom apps, shown at IFA 2024 (Berlin) and CES 2025 (Las Vegas)
- Built the 3D Map View, multi-site management and property dashboard for SmartThings Pro
- **Technical Skill:** Vue.js, Next.js, Three.js, Unity, Kotlin, Micronaut, Amazon Aurora, SSE, OAuth

**Senior Software Engineer**
***Samsung R&D Institute Bangladesh, Dhaka, Bangladesh*** **Mar 2022 - Mar 2025**

- Built a test management platform used by 35 QA engineers across 7 teams
- Developed features for an internal Atlassian Jira plugin
- **Technical Skill:** Java, React.js, Spring Boot, MySQL, MongoDB, Microservices, Jenkins, Nginx

**Software Engineer**
***Samsung R&D Institute Bangladesh, Dhaka, Bangladesh*** **Oct 2019 - Mar 2022**

- Built web applications and REST APIs presenting Samsung 5G network data through interactive views
- **Technical Skill:** Java, Spring Boot, Spring MVC, JPA, React, MySQL, MongoDB, Protocol Buffers

**Technical Content Writer**
***NSTU Code Academy (competitive programming blog)*** **Feb 2016 - Oct 2019**

- Wrote problem solutions and plain-language algorithm explanations: Blog

**Software Engineering Intern**
***Opus Technology Limited, Dhaka, Bangladesh*** **Aug 2018**

- Designed and built the base of an Android app with Java and Firebase Realtime Database

---

## ACHIEVEMENTS

**Excellence Award, Samsung R&D Institute Bangladesh (Oct - Dec 2024)**

- For work on Next SmartThings and Lounge, showcased at CES 2025, Las Vegas

**Icon Award, Samsung R&D Institute Bangladesh (May - Jun 2024)**

**Excellence Award, Samsung R&D Institute Bangladesh (Jan - Mar 2024)**

**Icon of the Month, Samsung R&D Institute Bangladesh (Mar - Apr 2022)**

- For major contributions to features of an Atlassian Jira plugin

**1st Runner-up, Intra University Programming Contest 2016, NSTU**

**2nd Runner-up, Intra ICE Programming Contest 2016, NSTU**

---

## LEADERSHIP AND SERVICE

- Vice President, ICE Programming Club, NSTU; mentored junior programmers
- Judge and problem setter: Intra ICE Programming Contest 2017, ICE FEST Programming Contest 2018, NSTU
- Problem setter: Intra University Programming Contest 2018 and 2019, NSTU

---

## CERTIFICATIONS

- Samsung Global Software Certificate, Professional Level, Samsung Electronics (Feb 2022)
- Samsung Global Software Certificate, Advanced Level, Samsung Electronics (Oct 2019)

---

## SKILLS

- **AI and ML:** LLM applications, agent workflows, LangChain, RAG, text-to-SQL, MCP / tool integration
- **Languages:** TypeScript, Java, Kotlin, Python
- **Frontend:** Vue.js, React, Next.js, Three.js, Unity 3D, Pinia, TanStack Query, Chart.js
- **Backend:** Spring Boot, Micronaut, Spring MVC, Spring Data / JPA, REST APIs, Microservices, SSE, OAuth
- **Data and Infrastructure:** PostgreSQL, MySQL, MongoDB, Amazon Aurora, Redis, RabbitMQ, Elasticsearch, Docker, Nginx, Jenkins, AWS